

IMO 2026 — Problem 2

Solution by Meta Muse Spark 1.1

2026-07-21

Source: `results/muse-spark-1.1/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

unsolved

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Approaches tried

- Setup coordinate system with midpoints at $(-1,0)$ and $(1,0)$ to exploit symmetry about perpendicular bisector of MN (which is y -axis). In this system, $OM=ON$ iff O 's x -coordinate is 0. Goal reduces to proving circumcenter of AKL lies on y -axis / line perpendicular to MN .
- Let $\alpha = KBA = ACL$, $\gamma = BMK = LCK$, $\beta = LBK = LNC$. Then angle $LBA = \alpha + \beta$, angle $ACK = \alpha + \gamma$. So lines determined. Parametrization leads to existence of K, L intersection. Trying to understand locus.
- Numeric search attempt: using triangle placing $M=(-1,0)$, $N=(1,0)$, $A=(u,v)$ etc. Then $B = (-2 -u, -v)$, $C = (2 -u, -v)$. Write equations for intersection. Need to solve for γ , β given α such that K and L definitions consistent.

Current best

Coordinate choice: $M=(-1,0)$, $N=(1,0)$, $A=(u,v)$. Then $B=(2M - A) = (-2 -u, -v)$, $C = (2N - A) = (2 -u, -v)$. BC horizontal line $y=-v$, MN horizontal $y=0$, etc. Condition $OM=ON$ equivalent to $Ox=0$ for circumcenter O of AKL. So need to prove $Ox=0$.

Let $\alpha = KBA = ACL$. Define $\gamma = BMK = LCK$, $\beta = LBK = LNC$.

Then geometry: - At B: BA direction to BM etc. But BA vector is $A - B$. BK is BA rotated clockwise by α (if inside). BL is BA rotated clockwise by $\alpha+\beta$. - At M: MB vector is $B - M$. MK is MB rotated CCW by γ . - At C: CA vector is $A - C$. CL is CA rotated??? Need orientation: $ACL = \alpha$, with L inside angle ACK. So CL is inside angle between CA and CK. CA direction from C to A. CK is further? Since L lies inside ACK, $ACL = \alpha$, and $ACK = \alpha + \gamma$? Let's deduce: $LCK = \gamma = BMK$. And $ACL = \alpha$, so $ACK = ACL + LCK = \alpha+\gamma$ if L inside ACK (with CL between CA and CK). So CK is CA rotated clockwise by $\alpha+\gamma$ (if CA as reference). Similarly at B: LBA? K inside angle LBA, $KBA=\alpha$, $LBK=\beta$, so $LBA=\alpha+\beta$. BA reference, BK is BA rotated (?) toward inside. So BL is further. - At N: NC vector = $C - N$. NL is? $LNC = \beta$, so angle between LN? Wait LNC vertex N between LN and CN. So angle between NL and NC is β . So NL is NC rotated.

Thus K is intersection of ray from B at angle α from BA and ray from M at angle γ from MB. So for each (α,γ) there is a K; consistency requires that C,K, and direction CA rotated by $\alpha+\gamma$ are collinear. That gives equation $E1(\alpha,\gamma)=0$. Similarly L intersection of ray from C at angle α from CA and ray from N at angle β from NC (?) gives condition that B,L, and BA rotated by $\alpha+\beta$ collinear: $E2(\alpha,\beta)=0$.

Thus K is fully determined by α (since γ solved from E1), L by α (β solved from E2). Subsequently need to show $Ox=0$.

We numerically suspect this holds.

Full proof

TBD.